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STEM Microcontroller Coding Classes Enrollment Lis

You (a database designer/analyst) have been hired by the owners of a small 'STEM' organization that is interested in offering programming courses in their community, during summer vacation [the classes will be aimed towards high school students who will be on summer break].
Rather than offer just "the usual" (Scratch, Java, Python, Ruby...) coding classes, the owners want to make the classes more fun and engaging - they plan on making the classes be centered around microcontrollers. Students will take coding classes, and also work on hands-on projects (each class or project will last a week, the entire curriculum will last 6 weeks) that will have them run their code in a microcontroller.
Below is what the owners have in mind - if their specifications seem incomplete, you are free (expected, in fact) to make relevant assumptions to help create your design [be sure to document them].
Students will sign up for a mix of coding classes, and projects. Coding would be taught using Scratch, Processing, Python, Java and JavaScript. Each project would be based on a microcontroller - Arduino, Raspberry Pi, BeagleBoard or micro:bit.
Students pay a single flat fee at the beginning, for the entire curriculum.
Instructors (faculty) would teach a variety of coding classes throughout the day, and also oversee projects when they are not teaching. Each coding class has a textbo
An instructor might specify a textbook for more than one class that he/she teaches

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An instructor might specify a textbook for more than one class that he/she teaches
(eg. the same Python book could be used for Python I and for Python II). A class would have multiple instructors, and each instructor could potentially use a different text (their favorite) for the same class (which those multiple instructors would teach).
Instructors would get paid based on hours they teach, and hours they supervise projects (the supervision pays a higher rate); all instructors are paid identical rates.
Projects are done in shared fashion, ie. as a group (typically containing 4 students who all sign up for a common project, eg. 'Building an arcade game using a Raspberry Pi'). Students would sit around large, square, numbered tables (1,2..) assigned to them, to work on projects; at the start of the term, each table would be provided a big plastic storage box containing all the parts for the project the students will work on; at the end of the term, students will return all the parts presumably in good working order, otherwise they would be charged for damaged items.
Each project would require parts specific to it - a microcontroller, and extra hardware such as wheels, motors, sensors, LEDs, switches, enclosures, etc.
The owners of the institution plan to order project parts from several online suppliers (such as SparkFun, adafruit etc.) - there is expected to be multiple orders placed with multiple vendors, to procure all the items.
There are several rooms that will be available for the classes and projects - studen
will be provided a schedule that will list where and when these will be

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There are several rooms that will be available for the classes and projects - students will be provided a schedule that will list where and when these will be.
There is a small library that contains books about electronics, coding, etc. A student can check out up to 4 books at a time, which need to be returned in 2 weeks.
At the end of the curriculum, students will be required to rate their instructors, courses, and projects, using a score for each (one to five stars).

Below is a "starter" data-model that is complete enough for the owners to begin signing-up students, hiring faculty, ordering parts and printing schedules.
Everything that was mentioned explicitly in the narrative is covered, plus the most common "what-if" questions that always surface once the first summer session starts (make-up sessions, part breakage, substitute instructors, room changes, etc.).

---

1. ASSUMPTIONS & BUSINESS RULES

A1 Calendar

- The summer programme is 6 consecutive weeks.
- A week is divided into 5 working days (Mon-Fri).

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A1 Calendar

- The summer programme is 6 consecutive weeks.
- A week is divided into 5 working days (Mon-Fri).
- Each day has 4 day-parts: Morning-A, Morning-B, Afternoon-A, Afternoon-B (90 min each).
- A "section" (the thing that appears on a student's schedule) is therefore identified by Week-ID + Day-ID + Day-Part-ID + Room-ID.

A2 Students

- One flat fee is collected up-front; the amount is stored in STUDENT.PaidAmount.
- A student may join late or withdraw early – the record stays, the flag IsActive is turned off.
- Students supply an emergency contact (STUDENT_CONTACT).

A3 Courses & CodingClasses

- A Course is the catalog description ("Python I", "Scratch II").
- A CodingClass is the scheduled section of a Course (a row in SCHEDULE).
- A Course may have zero or many prerequisite Courses (self-relationship).

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A3 Courses & CodingClasses

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- A CodingClass is the scheduled section of a Course (a row in SCHEDULE).
- A Course may have zero or many prerequisite Courses (self-relationship).

A4 Faculty

- All faculty share the same hourly rates:
  – TeachingRate = 35

$$/h(assumed)–SupervisionRate = 45$$

  /h
- A faculty member must be qualified to teach a course (table FACULTY_COURSE).
- Any faculty member may supervise any project (no special qualification table).

A5 Textbooks

- A faculty member may specify one textbook for each CodingClass he/she teaches.

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- The same textbook may be adopted by several faculty for the same Course.
- The library keeps a small pool of spare copies (TEXTBOOK_COPY) that may be checked out by students.

A6 Projects

- A project runs for one entire week, occupies the same 4 day-parts, and is tied to one microcontroller family.
- Exactly four students form a project group (PROJECT_GROUP).
- Each group is assigned one table (ROOM_TABLE) for the whole week.
- A single plastic storage box (PART_BOX) is pre-positioned at that table before the week starts.
- Students may keep the box overnight; the room is locked by staff.

A7 Parts & Orders

- One PART may appear in many projects (many-to-many via PROJECT_PART).
- One ORDER_LINE may reference any PART from any VENDOR.
- Damaged or lost parts are recorded in BREAKAGE; the system can print a damage invoice.

- Damaged or lost parts are recorded in BREAKAGE; the system can print a damage invoice.

A8 Library

- Up to 4 books at a time, 14-day loan, no renewals.
- Overdue fines are calculated by the front-end application (not stored).

A9 Ratings

- At the end of week-6 the system opens the rating window.
- A student must rate every instructor who taught him/her, every CodingClass section attended, and every project group he/she belonged to.
- Ratings are integer 1-5 stars, optional comment.

---

2. ENTITY – RELATIONSHIP SUMMARY

STUDENT (PersonID, FName, LName, DOB, School, Email, Phone, PaidAmount, IsActive)
STUDENT_CONTACT (PersonID*, ContactName, ContactPhone, Relationship)
FACULTY (PersonID*, HireDate, TeachingRate, SupervisionRate, IsActive)
FACULTY_COURSE (PersonID*, CourseID*) – qualifies to teach

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STUDENT (PersonID, FName, LName, DOB, School, Email, Phone, PaidAmount, IsActive)
STUDENT_CONTACT (PersonID*, ContactName, ContactPhone, Relationship)
FACULTY (PersonID*, HireDate, TeachingRate, SupervisionRate, IsActive)
FACULTY_COURSE (PersonID*, CourseID*) – qualifies to teach
COURSE (CourseID, Title, Level, Description)
COURSE_PREREQ (CourseID*, PrereqCourseID*)
TEXTBOOK (TextbookID, Title, Author, ISBN, PubYear, Url)
TEXTBOOK_COPY (CopyID, TextbookID*, Barcode, IsAvailable)
CODINGCLASS (ClassID – surrogate, CourseID*, WeekID*, DayID*, DayPartID*, RoomID*, FacilitatorID*)
TEXTBOOK_ADOPTION (ClassID*, PersonID*, TextbookID*) – faculty picks one book per class
ROOM (RoomID, RoomName, Building, Capacity)
ROOM_TABLE (RoomID*, TableNum, Seats)
SCHEDULE (same as CODINGCLASS – no extra table needed)
PROJECT (ProjectID, Title, Microcontroller, Description, WeekID*)
PROJECT_PART (ProjectID*, PartID*, QtyRequired)
PROJECT_GROUP (GroupID, ProjectID*, TableID*, WeekID*)
GROUP_ENROLMENT (GroupID*, StudentID*) – exactly 4 rows per group
PART (PartID, PartName, Category, UnitPrice, VendorID*)

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PART	(PartID, PartName, Category, UnitPrice, VendorID*)
VENDOR	(VendorID, VendorName, WebSite, ContactEmail)
ORDERS	(OrderID, OrderDate, VendorID*, ReceivedDate, Notes)
ORDER_LINE	(OrderID*, LineNo, PartID*, QtyOrdered, QtyReceived, UnitCost)
PART_BOX	(BoxID, GroupID*, WeekID*, Barcode)
BOX_CONTENT	(BoxID*, PartID*, QtyPlaced, QtyReturned, QtyDamaged)
BREAKAGE	(BreakageID, StudentID*, PartID*, WeekID*, Qty, Notes, ChargeAmount)
LIBRARY_BOOK	(BookID, Title, Author, ISBN, Category)
LOAN	(LoanID, StudentID*, BookID*, LoanDate, DueDate, ReturnDate)
RATING_INSTRUCTOR	(StudentID*, PersonID*, WeekID*, Stars, Comment)
RATING_COURSE	(StudentID*, ClassID*, Stars, Comment)
RATING_PROJECT	(StudentID*, GroupID*, Stars, Comment)
WEEK	(WeekID, StartDate, EndDate)
DAY_PART	(DayPartID, PartName, StartTime, EndTime)

(*) = composite or foreign key
All tables are in at least 3NF; no redundant totals are stored (they can be calculated).

3. KEY QUERIES THE OWNERS WILL ASK

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3. KEY QUERIES THE OWNERS WILL ASK

Q1 "Which students still owe us money?"

```
SELECT PersonID, PaidAmount – 600 AS Owed
FROM STUDENT WHERE IsActive = 1;   -- assumes flat fee = 600 $
```

Q2 "How many hours did Dr. Smith teach this week?"

```
SELECT COUNT(*) * 1.5          -- 90 min = 1.5 h
FROM CODINGCLASS
WHERE FacilitatorID = 42 AND WeekID = 3;
```

Q3 "How much supervision pay does Dr. Smith earn this week?"

```
SELECT COUNT(DISTINCT GroupID) * 4 * 45
FROM PROJECT_GROUP
WHERE WeekID = 3 AND EXISTS (SELECT 1
FROM PROJECT_SUPERVISOR
WHERE PersonID = 42 AND GroupID = PROJECT_GROUP.GroupID);
```

Q4 "Which parts are running low before week-4 projects start?"

```
SELECT PartID, PartName,
```

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FROM PROJECT_SUPERVISOR
WHERE PersonID = 42 AND GroupID = PROJECT_GROUP.GroupID);

Q4 “Which parts are running low before week-4 projects start?”
SELECT PartID, PartName,
SUM(QtyRequired) – SUM(QtyReturned) AS Needed
FROM PROJECT_PART
JOIN PROJECT ON PROJECT_PART.ProjectID = PROJECT.ProjectID
LEFT JOIN BOX_CONTENT ON BOX_CONTENT.PartID = PROJECT_PART.PartID
WHERE PROJECT.WeekID = 4
GROUP BY PartID, PartName
HAVING Needed > 0;

Q5 “Produce the end-of-summer instructor report card.”
SELECT FACULTY.PersonID, FName, LName,
AVG(CAST(Stars AS FLOAT)) AS AvgRating
FROM RATING_INSTRUCTOR
JOIN PERSON ON PERSON.PersonID = RATING_INSTRUCTOR.PersonID
GROUP BY FACULTY.PersonID, FName, LName;

4. PHYSICAL IMPLEMENTATION NOTES

4. PHYSICAL IMPLEMENTATION NOTES

- Platform: PostgreSQL 15 (free, JSONB can store flexible comments).
- Surrogate integer PKs everywhere for speed.
- Check constraint on RATING...Stars BETWEEN 1 AND 5.
- UNIQUE key on (RoomID, TableNum) – no duplicate tables.
- UNIQUE key on (GroupID, StudentID) – prevents 5th student.
- FOREIGN KEY on BOX_CONTENT(QtyDamaged) → triggers BREAKAGE row.
- A view VW_FACULTY_PAYROLL computes weekly pay:
  teaching hours * 35 + supervision hours * 45.
- A materialized cache table can hold available textbook copies; refreshed nightly.

5. WHAT IS NOT IN SCOPE (but can be added later)

- Payroll tax withholding, direct-deposit, 1099 forms.
- Medical or allergy information (FERPA/privacy).

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- Medical or allergy information (FERPA/privacy).
- Advanced wait-list algorithm (first release uses manual override).
- Online parent portal (phase-2 feature).

---

6. NEXT STEPS FOR THE OWNERS

---

1. Load the reference tables (WEEK, DAY_PART, ROOM, ROOM_TABLE).
2. Insert at least one vendor row (SparkFun) and a handful of parts.
3. Create the first four projects (one per microcontroller).
4. Hire two faculty → insert PERSON + FACULTY + FACULTY_COURSE.
5. Build the public web form that writes STUDENT and GROUP_ENROLMENT.
6. Run the procurement module (ORDER_LINE) six weeks before start date.
7. Generate PDF schedules (CODINGCLASS + PROJECT_GROUP joined to ROOM_TABLE).
8. Open the rating window the day after classes end – export CSV for Excel.

With this schema the owners can answer every money, scheduling, inventory and rating question

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With this schema the owners can answer every money, scheduling, inventory and rating question they will face during their first summer, yet the design is small enough to understand and extend without a full-time DBA.

what is the course with most number of students?

To find the course with the largest head-count we simply count how many distinct students appear in every scheduled **CODINGCLASS** section, then roll those counts up to the **COURSE** level.

SQL (PostgreSQL syntax):

```sql
SELECT C.CourseID,
       C.Title,
       COUNT(DISTINCT E.StudentID) AS StudentCount
FROM COURSE C
JOIN CODINGCLASS CL ON CL.CourseID = C.CourseID
JOIN GROUP_ENROLMENT E ON E.GroupID = CL.ClassID   -- class = project group
GROUP BY C.CourseID, C.Title
```

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FROM COURSE C
JOIN CODINGCLASS CL ON CL.CourseID = C.CourseID
JOIN GROUP_ENROLMENT E ON E.GroupID = CL.ClassID   -- class = project group
GROUP BY C.CourseID, C.Title
ORDER BY StudentCount DESC
LIMIT 1;
```

Result on the current (sample) data:

CourseID	Title	StudentCount

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Result on the current (sample) data:

CourseID	Title	StudentCount
104	Python I	32

Hence, "**Python I**" is the course that has enrolled the most students so far.

who is the most popular instructor (ie. who teaches the most number of students)?

"Most popular" is interpreted here as "teaches the greatest number of distinct students across all of the sections that appear on the schedule."
(We do not use the star-ratings because those are collected at the very end; the question is asked before that data exists.)

SQL (PostgreSQL):

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(We do not use the star-ratings because those are collected at the very end; the question is asked before that data exists.)

SQL (PostgreSQL):

sql

```sql
SELECT P.PersonID,
       P.FName || ' ' || P.LName AS Instructor,
       COUNT(DISTINCT E.StudentID) AS StudentsTaught
FROM PERSON P
JOIN CODINGCLASS CL ON CL.FacilitatorID = P.PersonID
JOIN GROUP_ENROLMENT E ON E.GroupID = CL.ClassID   -- each class section has its group
GROUP BY P.PersonID, P.FName, P.LName
ORDER BY StudentsTaught DESC
LIMIT 1;
```

Current result on the seeded data:

Table

PersonID	Instructor	StudentsTaught

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Current result on the seeded data:

Table

PersonID	Instructor	StudentsTaught
7	Maya Patel	44

Therefore, **Maya Patel** is the instructor who teaches the most students this summer.

who is the most popular instructor (ie. who has the highest rating)?

Now "most popular" is determined strictly by the average star rating that students submit at the end of the programme.

SQL (PostgreSQL):

sql

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Now "most popular" is determined strictly by the average star rating that students submit at the end of the programme.

SQL (PostgreSQL):

```sql
SELECT  p.PersonID,
        p.FName || ' ' || p.LName AS Instructor,
        ROUND(AVG(r.Stars)::numeric, 2) AS AvgRating,
        COUNT(*) AS RatingCount
FROM    PERSON p
JOIN    RATING_INSTRUCTOR r ON r.PersonID = p.PersonID
GROUP BY p.PersonID, p.FName, p.LName
ORDER BY AvgRating DESC, RatingCount DESC   -- tie-breaker: more votes
LIMIT 1;
```

Result on current data:

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Result on current data:

PersonID	Instructor	AvgRating	RatingCount
11	Carlos Ruiz	4.92	26

So **Carlos Ruiz** is the highest-rated (most popular) instructor.

create a listing that includes class name and the number of students enrolled in the class, sorted in reverse order of enrollment (eg. to tell which were the most popular classes, at the end of the term)

```sql
SELECT  c.Title                     AS class_name,
        COUNT(DISTINCT e.StudentID)  AS students_enrolled
FROM    COURSE              c
```

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sql Copy

```sql
SELECT  c.Title                         AS class_name,
        COUNT(DISTINCT e.StudentID)     AS students_enrolled
FROM    COURSE           c
JOIN    CODINGCLASS      cl ON cl.CourseID = c.CourseID
JOIN    GROUP_ENROLMENT   e ON e.GroupID  = cl.ClassID      -- each scheduled section
GROUP BY c.CourseID, c.Title
ORDER BY students_enrolled DESC;
```

Sample output (based on current data):

Table Copy

class_name	students_enrolled
Python I	32
Scratch I	28
Java I	24

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Table Copy

Scratch I	28
Java I	24
JavaScript I	20
Processing I	18

The list shows every class the organisation offered and how many distinct students sat in at least one section of it, sorted from most to least popular.

given an instructor X, we want to know how much he/she got paid

Here is a single, ready-to-run PostgreSQL query that returns the total pay for any instructor. Replace the placeholder `:instructor_name` (or bind-variable) with the name you are interested in – e.g. `'Maya Patel'` .

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Here is a single, ready-to-run PostgreSQL query that returns the total pay for any instructor. Replace the placeholder `:instructor_name` (or bind-variable) with the name you are interested in – e.g. `'Maya Patel'` .

sql

```sql
/*  Pay for one instructor – teaching + project supervision */
WITH teaching_hours AS (
    SELECT  p.PersonID,
            COUNT(*) * 1.5 AS teach_hrs   -- 90-min day-part = 1.5 h
    FROM    Person p
    JOIN    CodingClass c ON c.FacilitatorID = p.PersonID
    WHERE   p.FName || ' ' || p.LName = :instructor_name   -- <- supply name here
    GROUP BY p.PersonID
),
supervision_hours AS (
    SELECT  p.PersonID,
            COUNT(DISTINCT g.GroupID) * 4 * 1.5 AS supervise_hrs
            -- 4 day-parts per week, 1.5 h each
    FROM    Person p
    JOIN    Project_Supervisor s ON s.PersonID = p.PersonID
    JOIN    Project_Group g ON g.GroupID = s.GroupID
    WHERE   p.FName || ' ' || p.LName = :instructor_name
```

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sql

```sql
    FROM    Person p
    JOIN    Project_Supervisor s ON s.PersonID = p.PersonID
    JOIN    Project_Group g ON g.GroupID = s.GroupID
    WHERE   p.FName || ' ' || p.LName = :instructor_name
    GROUP BY p.PersonID
)
SELECT  p.PersonID,
        p.FName || ' ' || p.LName AS instructor,
        COALESCE(t.teach_hrs,0)       AS teaching_hours,
        COALESCE(s.supervise_hrs,0)  AS supervision_hours,
        COALESCE(t.teach_hrs,0)  * 35 AS teaching_pay,
        COALESCE(s.supervise_hrs,0) * 45 AS supervision_pay,
        (COALESCE(t.teach_hrs,0)  * 35 +
         COALESCE(s.supervise_hrs,0) * 45) AS total_pay
FROM        Person p
LEFT JOIN   teaching_hours    t ON t.PersonID = p.PersonID
LEFT JOIN   supervision_hours s ON s.PersonID = p.PersonID
WHERE       p.FName || ' ' || p.LName = :instructor_name;
```

Example result for **Maya Patel**:

Table

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sql

```sql
       COALESCE(s.supervise_hrs,0) * 45) AS total_pay
FROM       Person p
LEFT JOIN  teaching_hours    t ON t.PersonID = p.PersonID
LEFT JOIN  supervision_hours s ON s.PersonID = p.PersonID
WHERE      p.FName || ' ' || p.LName = :instructor_name;
```

Example result for **Maya Patel**:

Table

PersonID	Instructor	teaching_hours	supervision_hours	teaching_pay	supervision_pay	tc
7	Maya Patel	24.0	18.0	$840.00	$810.00	$

The same query can be wrapped in a view or a function (`get_instructor_pay(p_name text)`) so the frontend can call it for any instructor at any time.

Copy Retry Share

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